

# Castro R S Jeev

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## PROFESSIONAL SUMMARY:

AI Engineer with hands-on experience building and deploying production AI systems — a hybrid-search RAG pipeline with citation verification, a Teams-integrated meeting bot, and a 79-class ASL sign language recognition model. Comfortable across the full stack: LangChain, FAISS, Pinecone, PyTorch. Looking for an AI Engineer / Machine Learning Engineer role where I can keep shipping things like this.

## EDUCATION:

**B.E. Computer Science & Engineering (Honours in Data Science) — Mar Ephraem College of Engineering & Technology**

CGPA: 8.89

## TECHNICAL SKILLS:

**Language:** Python, JavaScript

**Generative AI & LLMs:** Langchain, Prompt Engineering

**Vector Search & Retrieval:** Pinecone, FAISS, Hybrid Search, RRF, Semantic Chunking

**Machine Learning & Deep Learning:** PyTorch, Scikit-Learn, CNNs, RNNs, LSTMs, GRUs, XGBoost

**Computer Vision:** OpenCV, Mediapipe

**Web & Deployment:** FastAPI, REST APIs, Streamlit

**Evaluation:** NLI Based Verification, Evaluation Metrics, RAGAS

**Tools & Deployment:** Git/Github, Huggingface

## EXPERIENCE:

**AI Intern – CodeWork AI, Chennai, India (onsite)**

**Jul 2026- Present**

### Bob bot for MS Teams:

*Tech: Bot Framework, ACS Call Automation, Microsoft Graph API, Node.js, Python*

- Built a bot for Microsoft Teams that lets technical managers join meetings on their behalf, using the Graph API to enter calls, speak provided content, and listen to the conversation
- Answers questions asked during the meeting by retrieving relevant documents in real time

### Sign Language Translation (ASL):

*Tech: MediaPipe Holistic, PyTorch, Conv1D + GRU, ONNX, PySide6*

- Researched and built a sign language recognition model for American Sign Language (ASL), covering 79 signs
- Collected and cleaned ASL video data from sources with commercial-use licensing, then extracted hand, face and body landmarks using MediaPipe Holistic.
- Researched and built a custom Conv1D + GRU architecture trained on the landmark features, later improving accuracy with ArcFace metric learning.
- The final model covers 79 signs with ~89% validation accuracy (~90.4% with ArcFace)

## PROJECTS:

### Citation Accurate RAG for Research Paper PDFs:

[GITHUB](#) | [LIVE](#)

*Tech: Pinecone, PyMuPDF, Groq, Hybrid Search (BM25 + Dense), RRF*

- Built a RAG pipeline for querying research paper PDFs, using PyMuPDF for text extraction and Pinecone as the vector database and embedding provider.
- Used GROQ's llama model as main LLM with forced json output. Implemented hybrid retrieval using Sparse and Dense vector search.
- Used Reciprocal Rank Fusion for merging the chunks retrieved by hybrid search

### AI Interview Evaluation Agent:

[GITHUB](#) | [LIVE](#)

*Tech: LangChain, PyMuPDF, Gemini, Groq, Ollama, FAISS, Tavily*

- Built a multi-stage interview evaluation agent using LangChain, with resume parsing via PyMuPDF and Gemini's multimodal model
- Set up a fallback chain across Gemini, Groq and Ollama so the agent keeps working even if one LLM provider goes down
- Added FAISS-based retrieval and Tavily web search to ground answers instead of relying purely on the LLM
- Built a structured answer-evaluation pipeline and deployed the app on Render